REVA UNIVERSITY
Bengaluru, India

School of Computer Science & Engineering
AI Application Development – IA1

FarmTrace

Team members & roles

TEAM MEMBER NAME	ROLE
NISHA YADAV	TEAM LEAD
SANJANA R	DEVELOPER
NAMEERA SANIYA	GITHUB/PRODUCT BUILD
RUHI SHARMA	DEVELOPMENT LEAD
MEGHANA U	TEST LEAD
SAGARIKA DS	CUSTOMER CONTACT LEAD

Modified Problem Statement

Small-scale farmers in India cannot prove the origin or quality of their produce to buyers, resulting in lost income, broken trust, and limited market access due to the absence of affordable traceability tools.

User/ Customer Understanding

We interviewed 5 potential customers — 4 small-scale farmers and 2 vegetable buyers (Bengaluru) — and photographed each session. Photos available in GitHub documents folder.

Ideal Customer Profile (ICP)

Demographic Profile:
Small-scale farmer, age 25–55, owning 1–5 acres in Karnataka .
Annual income Rs.1–3 lakhs.
Basic Android smartphone user.
Sells via local mandi or aggregators.

Psychographic Profile:
Proud of produce quality but frustrated by low prices.
Practices organic/natural farming but cannot prove it to buyers.
Motivated by fair income.
Open to simple digital tools that show immediate income benefit.
Distrusts middlemen.

Customer Persona (CP)

Name: Ravi Kumar (fictitious) | Age: 38 | Location: Kolar, Karnataka
Demographic: 3-acre tomato & chilli farmer. 10th standard education. Android user (WhatsApp). Sells to local aggregator. Annual income ~Rs.1.8 lakhs.

Psychographic: Hardworking and quality-conscious. Frustrated that buyers don’t trust his organic claims. Gets Rs.8/kg when organic shops sell the same tomatoes for Rs.14/kg. Would use FarmTrace if it helps him prove quality and earn a fair price.

Product USP

- Zero cost to farmer runs entirely on free-tier cloud; no hardware or subscription fees unlike expensive blockchain traceability systems

- No app download needed buyer scans QR with any phone camera and sees full product history instantly in a browser

- Farmer registers a batch in under 2 minutes designed for low digital literacy; works on any basic Android phone

- Provides verifiable proof of organic farming and origin helps farmers earn 15–20% higher prices by building buyer trust

Use Cases & High level Requirements

UC1: Farmer registers batch — Input: crop name, village, harvest date, method | Output: unique QR code generated instantly
UC2: Buyer scans QR — Input: phone camera scan | Output: full product page (origin, date, method, timeline) in browser — no app needed
UC3: Aggregator adds handling event — Input: batch ID + event description + date | Output: updated journey timeline visible to all buyers
UC4: Farmer shares QR on WhatsApp — Input: screenshot of QR | Output: buyer taps link, views product history, places order with confidence
UC5: Exporter verifies organic batch — Input: QR scan on received crate | Output: confirmed organic method + farm origin — approves premium pricing

Requirements document available on GitHub — includes Functional Requirements , Non-Functional Requirements (performance, cost, usability), and Acceptance Criteria for each feature.

Proof of Concept (PoC) / Prototype Demo

https://github.com/Sanjanarprasad/Supply_chain/blob/main/docs/demo.mp4